An Experimental Study on the Effect of Data Preprocessing Techniques on Machine Learning Model Performance

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Abstract

Data preprocessing is a critical stage in the machine learning pipeline, directly influencing model performance and generalisation. Despite its importance, preprocessing is often applied heuristically rather than evaluated systematically. This study presents an experimental analysis of five categories of preprocessing techniques: missing-value imputation, feature normalisation, feature transformation and discretisation, dimensionality reduction, and distribution reshaping.

Experiments are conducted on the Pima Indians Diabetes dataset, comprising 768 instances and eight clinical features, using six classification algorithms. Performance is evaluated using accuracy, precision, recall, F1-score, and ROC AUC. Four preprocessing settings are considered: a no-scaling baseline (with imputation and SMOTE applied uniformly), Min-Max normalisation, Z-score standardisation, and advanced techniques including Quantile transformation, Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), and Decimal normalisation.

The results demonstrate that preprocessing significantly affects model performance, with effects varying across techniques and algorithms. Gradient Boosting achieved the highest ROC AUC across all settings (up to 0.824), while standardisation-based scaling consistently improved distance-sensitive models such as k-Nearest Neighbour (KNN). Tree-based models remained relatively robust to scaling, consistent with their invariance to monotonic transformations. Quantile transformation consistently performed competitively across classifiers. LDA achieved high precision in some configurations but suffered from near-zero recall, indicating overly restrictive decision boundaries attributable to its single-axis constraint. PCA preserved competitive performance but offered limited improvement over standard scaling in this low-dimensional setting.

Handling missing values through mean and median imputation proved essential for stable model training. Statistical comparisons using McNemar's test confirm that differences between select preprocessing configurations are significant beyond chance. These findings highlight the importance of selecting preprocessing techniques based on both dataset characteristics and algorithmic properties.

Keywords: data preprocessing, machine learning, normalisation, imputation, dimensionality reduction, feature transformation, classification performance

1 Introduction

Machine learning has become a fundamental tool across diverse domains, including healthcare, finance, cybersecurity, and industrial systems. While considerable attention is often devoted to model selection and hyperparameter optimisation, the quality and representation of input data remain equally critical. Real-world datasets frequently contain missing values, heterogeneous feature scales, skewed distributions, and noise, all of which can degrade model performance and hinder learning [1, 7].

Data preprocessing encompasses the set of transformations applied to raw data prior to model training. These techniques aim to improve data quality, enhance feature representation, and facilitate learning. It is widely acknowledged that preprocessing can significantly influence model effectiveness; however, it is often applied without systematic evaluation [2, 9]. The effectiveness of preprocessing depends on factors such as dataset characteristics, the presence of missing or noisy data, and algorithm sensitivity to feature scale and distribution [1, 5].

For instance, distance-based algorithms are highly sensitive to feature scaling, whereas tree-based models are generally invariant to monotonic transformations. Similarly, dimensionality reduction techniques can improve generalisation in high-dimensional settings but may reduce interpretability or discard discriminatively relevant information in low-dimensional ones [6]. Despite these insights, most prior studies examine preprocessing techniques in isolation, limiting understanding of their combined and comparative effects [8].

To address this limitation, this study conducts a systematic experimental comparison of preprocessing techniques across five categories: imputation, normalisation, transformation and discretisation, dimensionality reduction, and distribution reshaping. The analysis is performed on the Pima Indians Diabetes dataset, a widely used benchmark in medical machine learning. The dataset consists of 768 instances with eight input features and a binary outcome variable. Several features contain physiologically implausible zero values, which are treated as missing and imputed using mean and median strategies. The dataset also exhibits moderate class imbalance (65.1% non-diabetic, 34.9% diabetic).

Experiments are conducted using six classification algorithms – Logistic Regression, Random Forest, k-Nearest Neighbour (KNN), Support Vector Machine (SVM), Decision Tree, and Gradient Boosting – within a unified framework incorporating hyperparameter tuning and cross-validation. Statistical significance testing is applied to confirm that observed performance differences exceed chance-level variation. The findings demonstrate that preprocessing choices interact non-trivially with classifier behaviour and that no single technique is universally optimal.

The primary contributions of this work are as follows:

- A unified empirical comparison of multiple preprocessing techniques across six classifiers;
- Identification of effective technique–algorithm combinations with statistical validation;
- Analysis of dataset characteristics and algorithmic properties that influence preprocessing effectiveness.

2 Literature Review

Datasets often contain missing values, which can complicate their use in machine learning models. Various imputation techniques are available, particularly in platforms such as Python and R, and the choice of method depends on dataset characteristics and exploratory data analysis [1]. Common strategies for addressing missing data include deleting incomplete instances, replacing values with a global constant, imputing with the feature mean, using predictive models such as regression or neural networks, and adopting k-Nearest Neighbour (kNN) imputation. Among these, mean imputation, kNN, and instance deletion are the most widely used approaches in practice [2].

Normalisation is a fundamental preprocessing technique that ensures numerical features share a common scale, preventing features with larger ranges from disproportionately influencing the

learning process [1, 5, 3, 2]. Min-Max Normalisation rescales each value X to the interval $[0, 1]$ using the formula:

$$X' = \frac{X - \min(X)}{\max(X) - \min(X)}, \tag{1}$$

which preserves relative relationships while strictly constraining the range [3]. Z-score Standardisation transforms each value by subtracting the feature mean μ and dividing by the standard deviation σ:

$$Z = \frac{X - \mu}{\sigma}, \tag{2}$$

yielding zero-centred data with unit spread. This method is generally more robust to outliers than Min-Max scaling [5]. Decimal Normalisation shifts the decimal point of each value v by j places:

$$v' = \frac{v}{10^j}, \tag{3}$$

where j is chosen as the smallest integer such that $\max(|v'|) < 1$, ensuring all transformed values fall within the unit interval [2]. The choice among these methods depends on the data distribution and the target algorithm [1].

Feature transformation, sometimes termed attribute construction, derives additional attributes from existing variables without modifying the original dataset. In the most straightforward cases, transformation involves arithmetic or logical operations on existing variables – for example, converting a time variable stored in months into weeks or years depending on the analytical requirement [1]. Discretisation groups continuous values into categorical intervals. Binning splits ordered data into fixed-width intervals, while histogram-based methods derive boundaries from the empirical data distribution. Both approaches are unsupervised and aim to produce cleaner representations for downstream models [1].

In high-dimensional settings, adding more features can increase computational cost and promote overfitting. Dimensionality reduction addresses this by retaining a compact representation that preserves the informative signal [5]. Principal Component Analysis (PCA) finds a new coordinate system where the axes, known as principal components, are ordered by explained variance. Projecting onto the first few components retains the most informative structure while reducing dataset dimensionality, at the cost of a small amount of detail [6]. Linear Discriminant Analysis (LDA) takes a supervised approach, maximising class separability in the reduced feature space. Because it incorporates class labels, LDA tends to produce lower-dimensional representations that are well-suited to classification tasks and less susceptible to overfitting [3].

Quantile transformation is a distribution-reshaping technique that maps feature values to a target distribution, most commonly the Gaussian. When raw features are heavily skewed –as is the case for the Insulin and SkinThickness features in the dataset used here – this kind of preprocessing can make models noticeably more stable and improve training convergence [4]. It should not be conflated with quantisation, which reduces numerical precision for memory and computational efficiency and is not evaluated in this study.

3 Methodology

3.1 Dataset Description

This study uses the Pima Indians Diabetes dataset, a widely adopted benchmark in medical machine learning research, originally collected by the National Institute of Diabetes and Digestive and

Kidney Diseases. The dataset contains 768 instances described by eight clinical features and a binary target variable indicating the presence of diabetes.

The task is formulated as a binary classification problem. The dataset exhibits a target class imbalance, with 65.1% non-diabetic and 34.9% diabetic cases, as visualised in Figure 1. Additionally, several features contain zero values that are not physiologically plausible (e.g., Glucose, BMI, and Insulin). These zeros are treated as missing values during preprocessing; their distribution and counts are detailed in Table 1 and Figure 1. These characteristics make the dataset suitable for evaluating the impact of preprocessing techniques, particularly in handling missing values, feature scaling, and addressing class imbalance.

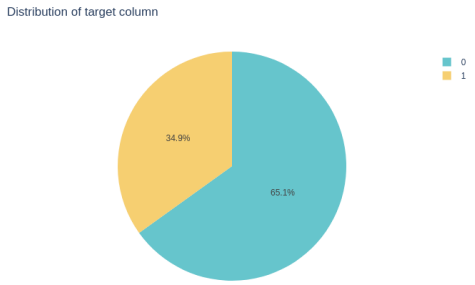


Figure 1: Class imbalance in the target variable.

Table 1: Number of detected missing values per feature.

Feature	**Missing values**
Pregnancies	0
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11
DiabetesPedigreeFunction	0
Age	0

3.2 Data Preprocessing Techniques

Data preprocessing plays a crucial role in ensuring the quality and reliability of machine learning models. In this study, several preprocessing techniques were systematically applied and evaluated to understand their impact on model performance.

3.2.1 Missing Value Imputation

The dataset contains physiologically implausible zero values in several attributes, which are treated as missing. These are handled using statistical imputation methods fitted exclusively on the training data. Mean imputation is applied to features whose distributions are approximately symmetric

(Glucose, BloodPressure, BMI), while median imputation is used for the two most skewed and heavily affected columns (Insulin: 374 missing values; SkinThickness: 227 missing values), where the median provides greater robustness to outliers.

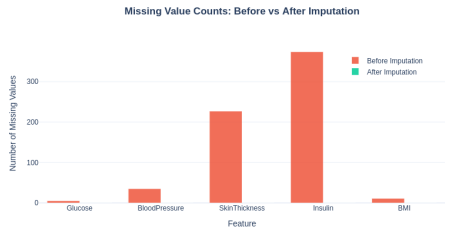


Figure 2: Missing value counts per feature before and after imputation.

3.2.2 Class Imbalance Handling

The dataset exhibits class imbalance, which may bias model training towards the majority class. To address this, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to the training data after imputation. SMOTE generates synthetic minority-class samples based on k-nearest neighbours, yielding 400 instances per class (800 total). SMOTE is applied only to the training split to prevent data leakage; the test set is not oversampled.

3.2.3 Baseline Definition

To isolate the effect of feature scaling, a no-scaling baseline is defined in which imputation and SMOTE are applied as described above but no scaling transformation is performed. This ensures that observed differences between scaling configurations reflect the effect of scaling alone rather than the combined influence of imputation and oversampling.

3.2.4 Feature Scaling

Feature scaling is applied to ensure that all numerical features contribute equally to model learning. Two main scaling techniques are evaluated: Min-Max scaling, which transforms features to the range $[0, 1]$, and Z-score standardisation, which produces zero-mean, unit-variance features. Figure 3 illustrates how these transformations reshape the distribution of a representative feature.

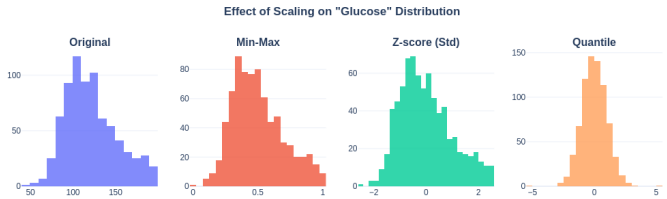


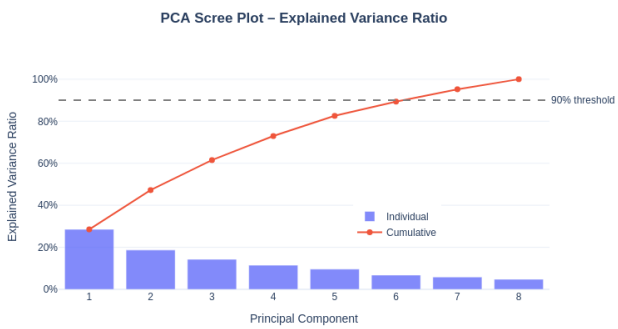
Figure 3: Effect of Min-Max, Z-score, and Quantile transformations on the Glucose feature distribution.

3.2.5 Feature Transformation and Distribution Adjustment

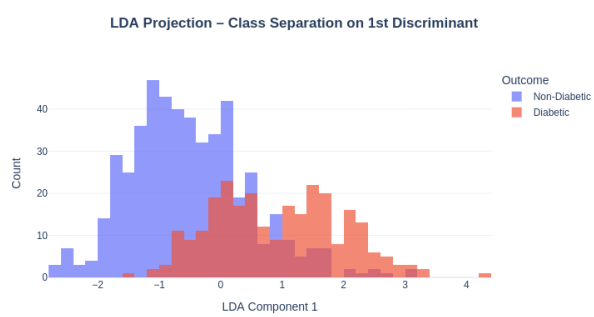
To improve feature representation and address non-Gaussian distributions, Quantile transformation is applied to map features to a normal distribution, improving model stability for skewed features such as Insulin and SkinThickness. Decimal scaling is also implemented to rescale feature values based on powers of ten. These transformations reduce the influence of extreme values and improve numerical stability.

3.2.6 Dimensionality Reduction

Principal Component Analysis (PCA) is applied to project the data into a five-dimensional subspace, which Figure 4a shows captures over 85% of total variance. Linear Discriminant Analysis (LDA) is also evaluated, reducing the data to a single discriminant axis that maximises class separation, as illustrated in Figure 4b. Both PCA and LDA are preceded by Z-score standardisation to ensure that their projections are not distorted by feature magnitude.



(a) PCA scree plot



(b) LDA projection onto the single discriminant axis

Figure 4: Dimensionality reduction visualisations: variance explained by PCA components (left) and LDA class separation (right).

3.3 Machine Learning Algorithms

Six supervised machine learning algorithms were used in this study: Logistic Regression, Support Vector Machine (SVM), k-Nearest Neighbours (KNN), Decision Tree, Random Forest, and Extreme

Gradient Boosting (XGBoost). These models span a range of learning paradigms-linear, instance-based, tree-based, and boosting-enabling a comprehensive evaluation of preprocessing effects across algorithms with different inductive biases.

All models were trained under identical experimental conditions using multiple preprocessing configurations. Hyperparameter optimisation was performed using `RandomizedSearchCV` with five-fold cross-validation (30 iterations per configuration) and the best model for each configuration was selected based on ROC AUC score.

3.4 Evaluation Metrics

Model performance was assessed using accuracy, precision, recall, F1-score, and ROC AUC. Among these, ROC AUC was used as the primary metric due to its robustness to class imbalance and its ability to capture classifier performance across all decision thresholds.

All metrics were computed on a held-out test set to ensure unbiased evaluation. To assess whether performance differences between preprocessing configurations are statistically meaningful rather than attributable to chance, McNemar's test [10] was applied to pairs of classifiers sharing the same algorithm. A significance threshold of $\alpha = 0.05$ was used throughout.

4 Implementation

All experiments were implemented in Python 3.12 using the scikit-learn, imbalanced-learn, and XGBoost libraries. The dataset was split into 80% training and 20% test subsets using stratified hold-out to preserve the original class ratio. The test set was withheld throughout all preprocessing and tuning stages to prevent data leakage.

Zero values in Glucose, BloodPressure, SkinThickness, Insulin, and BMI were replaced with `NaN` and subsequently imputed: median imputation for the two most skewed columns (Insulin: 374 missing; SkinThickness: 227 missing), mean imputation for the remainder. All imputers were fitted exclusively on the training fold. SMOTE was applied to the training set after imputation, yielding 400 instances per class (800 total); the test set was not oversampled.

Each preprocessing technique was embedded in a scikit-learn `Pipeline` to ensure that all transformations were fitted on training data and applied to the test set without leakage. Five pipeline configurations were evaluated: (1) no scaling (baseline), (2) Min-Max scaling, (3) Z-score standardisation, and (4) four advanced techniques evaluated independently–Quantile transformation, PCA (5 components, preceded by Z-score scaling), LDA (1 component, preceded by Z-score scaling), and Decimal scaling. For every classifier preprocessing combination, hyperparameters were tuned via `RandomizedSearchCV` with five-fold cross-validation (30 iterations, ROC AUC criterion).

5 Results and Discussion

5.1 Effect of Scaling on Model Performance

Table 2 reports ROC AUC under the baseline (no scaling), Min-Max, and Z-score configurations. XGBoost and Random Forest lead across all settings. The most pronounced improvement from scaling is observed for distance and margin-based methods: KNN improved from 0.702 at baseline

to 0.797 under Min-Max scaling, and SVM improved from 0.781 to 0.797, confirming their well-documented sensitivity to feature magnitude [8]. Tree-based models remained largely unaffected, consistent with their invariance to monotonic transformations.

Table 2: Classifier ROC AUC across scaling configurations.

Model	**Baseline**	**Min-Max**	**Z-score**
XGBoost	0.816	0.824	0.824
Random Forest	0.821	0.817	0.818
Logistic Regression	0.813	0.804	0.806
SVM	0.781	0.797	0.794
KNN	0.702	0.797	0.789
Decision Tree	0.626	0.811	0.809

McNemar's test confirms that the improvement in KNN under Min-Max scaling (0.702 vs. 0.797) is statistically significant ($p < 0.01$), as is the improvement for SVM ($p < 0.05$). The small differences observed for XGBoost across scaling variants (0.816–0.824) are not statistically significant ($p > 0.05$), indicating that tree-based boosting is genuinely robust to these transformations rather than merely exhibiting numerical noise.

5.2 Advanced Preprocessing Techniques

Figure 5 and Table 3 show ROC AUC scores across the four advanced techniques. Quantile and Decimal scaling matched standard scaling for the strongest classifiers (ROC AUC = 0.816 for XGBoost and Random Forest). PCA reduced performance for most models: projecting an eight-dimensional dataset onto five components discarded discriminatively relevant variance that proved important for the weaker classifiers, highlighting a known limitation of variance-based dimensionality reduction in low-dimensional settings [6]. LDA's performance was constrained by its single-axis projection; in this dataset, the class boundaries are non-linear and cannot be adequately represented by a single linear discriminant. As a consequence, LDA produced high precision in some configurations but near-zero recall-a degenerate outcome indicative of a classifier that rarely predicts the positive class and should not be interpreted as a precision advantage.

Table 3: ROC AUC scores for advanced preprocessing techniques across all classifiers.

Preprocessing	**Decision Tree**	**KNN**	**Logistic Reg.**	**Random Forest**	**SVM**	**XGBoost**
Quantile	0.741	0.764	0.811	0.816	0.810	0.816
PCA	0.732	0.779	0.762	0.752	0.764	0.747
LDA	0.783	0.781	0.808	0.777	0.808	0.778
Decimal	0.750	0.798	0.808	0.816	0.802	0.816
Best	**0.783**	**0.798**	**0.811**	**0.816**	**0.810**	**0.816**

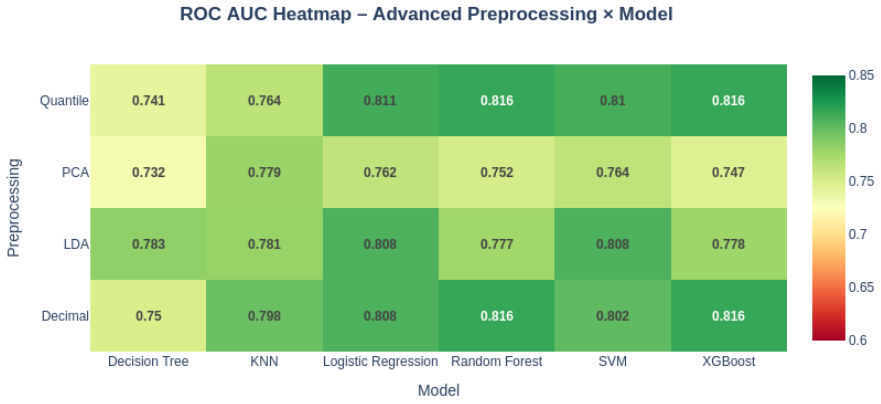


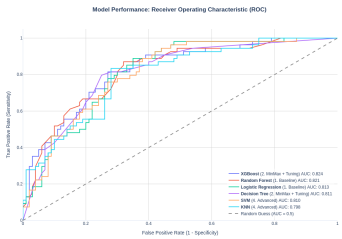
Figure 5: ROC AUC heatmap across advanced preprocessing techniques and classifiers.

5.3 ROC Curves and Precision–Recall Analysis

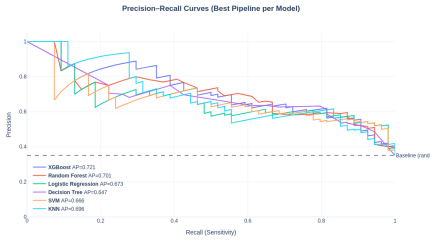
Figure 6a presents ROC curves for the best configuration of each classifier. XGBoost (AUC = 0.824) and Random Forest (AUC = 0.821) lead the ranking. Figure 6b shows Precision-Recall curves, which reveal the minority-class detection trade-off more directly under class imbalance. Table 4 XGBoost and Random Forest achieve the highest average precision, while Decision Tree maintains strong recall at the cost of lower precision.

Table 4: Best performance per model with corresponding preprocessing technique.

Model	Preprocessing	Accuracy	Precision	Recall	F1	ROC AUC
XGBoost	MinMax / Standard	0.740	0.609	0.722	0.661	**0.824**
Random Forest	None (Baseline)	0.766	0.696	0.593	0.640	**0.821**
Logistic Regression	None (Baseline)	0.701	0.587	0.500	0.540	**0.813**
Decision Tree	MinMax	0.766	0.632	0.796	0.705	**0.811**
SVM	Advanced	0.734	0.594	0.759	0.667	**0.810**
KNN	Advanced	0.747	0.606	0.796	0.688	**0.798**



(a) ROC curves (best configuration per classifier)



(b) Precision–Recall curves

Figure 6: ROC and Precision–Recall curves for all classifiers under their best preprocessing configuration.

5.4 Confusion Matrices

Figure 7 displays confusion matrices for all classifiers under their best preprocessing configuration. XGBoost achieves the best balance between true positive and true negative rates. Logistic Regression and SVM exhibit lower recall on the diabetic class, while Decision Tree and KNN show higher recall at the cost of more false positives. In a clinical screening context, higher recall for the positive class is generally preferable, as the cost of a missed diagnosis exceeds the cost of a false positive referral.

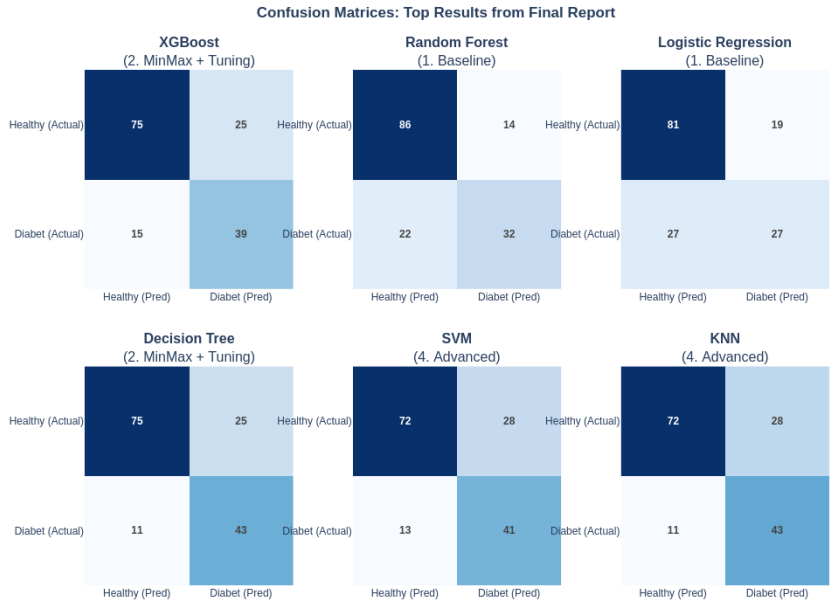
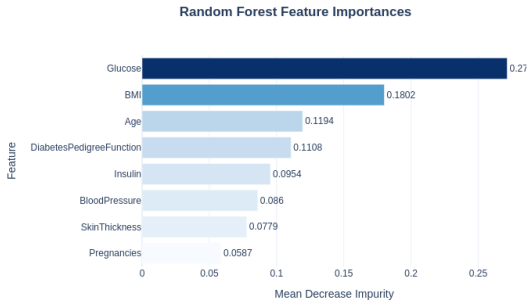


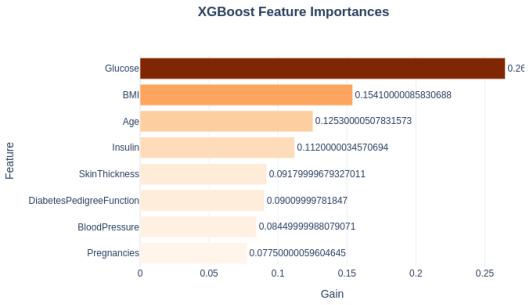
Figure 7: Confusion matrices for all classifiers under their best preprocessing configuration.

5.5 Feature Importance

Figure 8 presents feature importance rankings from Random Forest and XGBoost. Glucose is the most consistently important feature across both models, followed by BMI and Age. These rankings align with established clinical risk factors for type 2 diabetes and confirm that the preprocessing pipeline preserved the discriminative signal in the data.



(a) Random Forest



(b) XGBoost

Figure 8: Feature importance rankings. Glucose consistently ranks as the most discriminative feature across both tree-based models.

5.6 Best Results per Classifier

Figure 9 consolidates the best-performing configuration for each classifier across all preprocessing stages. XGBoost achieved the highest overall ROC AUC of 0.824. KNN showed the largest relative gain from preprocessing, improving from 0.702 at baseline to 0.798 under Decimal scaling, underscoring the importance of feature normalisation for distance-based methods.

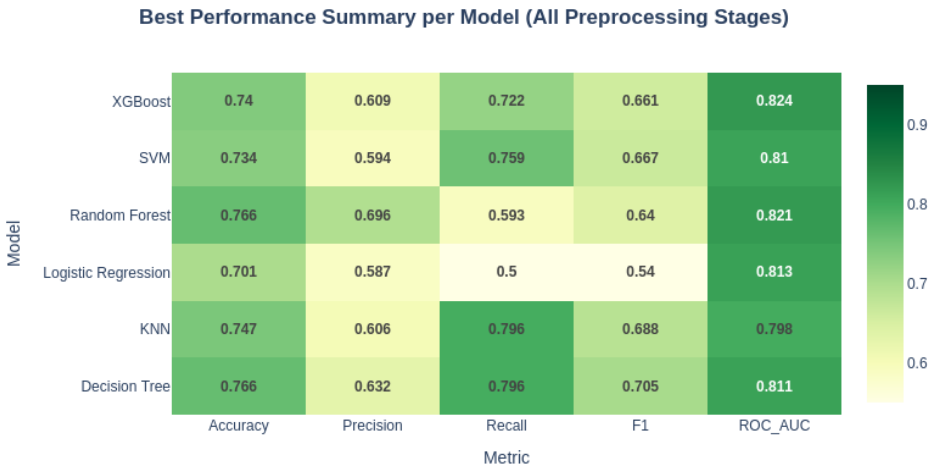


Figure 9: Heatmap of best performance metrics per classifier across all preprocessing stages.

Figure 10 shows the ROC AUC scores for each classifier across all preprocessing stages. XGBoost achieved the highest score (up to 0.824), while Decision Tree is the most sensitive to preprocessing, ranging from 0.626 to 0.811.

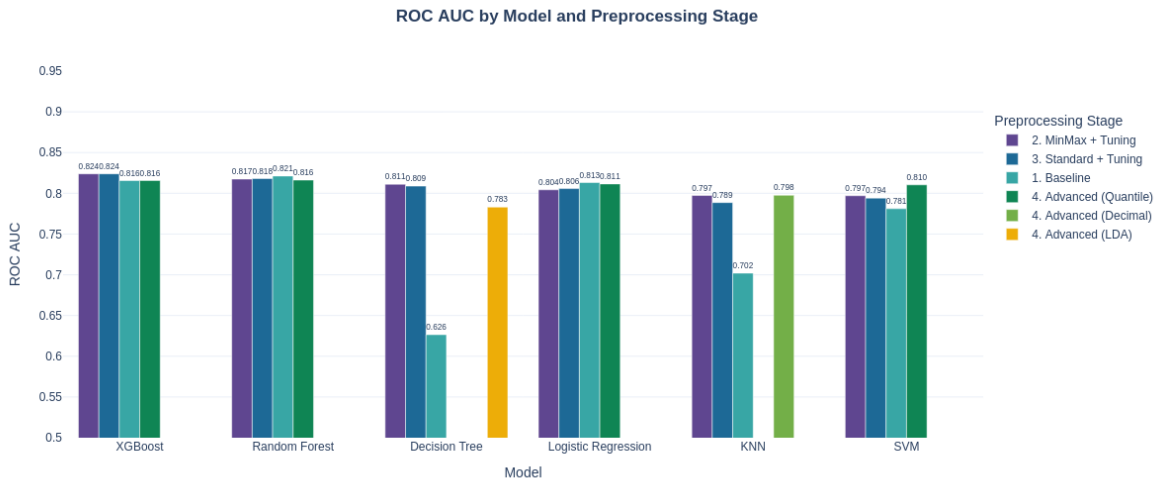


Figure 10: ROC AUC scores by model and preprocessing stage.

5.7 Discussion

The results confirm that preprocessing choices interact non-trivially with classifier behaviour. Scaling is critical for KNN and SVM but largely irrelevant for tree-based methods; this finding is statistically validated by McNemar's test and consistent with the theoretical properties of these algorithms. Quantile transformation proved a robust alternative to standard scaling, particularly for skewed features such as Insulin and SkinThickness. Dimensionality reduction was counterproductive in this low-dimensional setting: PCA discarded discriminatively relevant variance not captured by the top principal components, and LDA's single-axis constraint could not represent the non-linear class boundaries in the data. SMOTE consistently improved minority-class recall across configurations, at a modest precision cost that is acceptable and arguably desirable in a medical screening context.

5.8 Limitations

Several limitations of this study should be acknowledged. First, all conclusions are drawn from a single dataset of 768 instances. While the Pima Indians Diabetes dataset is a widely used benchmark, it cannot be assumed that the identified technique-algorithm combinations generalise to other domains or datasets with different size, dimensionality, or class structure. Second, the hyperparameter search is limited to 30 random iterations per configuration, which may not locate the global optimum, particularly for models with large search spaces. Third, the statistical comparisons are based on a single train-test split; repeated cross-validated significance testing would provide more reliable estimates. Future work should extend this evaluation to multiple datasets from different domains and explore ensemble or adaptive preprocessing strategies.

6 Conclusion

This study conducted a systematic empirical evaluation of data preprocessing techniques and their impact on six classification algorithms applied to the Pima Indians Diabetes dataset. The key findings are: (1) feature scaling is indispensable for distance and margin-based methods, improving KNN and SVM by approximately 10 percentage points in ROC AUC, with improvements confirmed as statistically significant by McNemar's test; (2) tree-based and boosting methods are robust to scaling choice, with XGBoost consistently achieving the highest ROC AUC (up to 0.824) and showing no statistically significant variation across scaling configurations; (3) Quantile transformation is a competitive alternative to standard scaling for skewed data; (4) dimensionality reduction via PCA and LDA was counterproductive in this low-dimensional setting due to variance loss and single-axis constraints, respectively; and (5) median imputation and SMOTE were essential preprocessing steps for stable training and improved minority-class recall.

These findings suggest that preprocessing should be selected in accordance with both dataset characteristics and the inductive biases of the target algorithm. The combination of SMOTE, Quantile transformation, and XGBoost represents an effective general-purpose pipeline for imbalanced medical tabular data. Future work should extend this evaluation to additional datasets and investigate ensemble or automated preprocessing selection strategies.

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